# AI Sports Analyst System

An Object Detection and Tracking Pipeline Using YOLOv8 and DeepSORT

Project Report




Figure 1. YOLOv8 visual identity used in the project materials.

**Prepared from the supplied project materials**

The report documents the system design, implementation, evaluation outputs, limitations, and future work.

# Executive Summary

This project presents an AI Sports Analyst System designed to detect and track important objects in football video and images. The system combines YOLOv8 for object detection with DeepSORT for multi-object tracking and uses a Gradio dashboard to provide an interactive interface for uploading sports images and viewing detection results. The supplied outputs show detections of players and referees, confidence scores, a confusion matrix, and dashboard-level interaction. The project demonstrates how computer vision can transform raw sports footage into structured observations that support analysis, review, and future statistical services.

The report follows the required structure supplied with the project brief. Where exact dataset counts, training hyperparameters, or numerical precision/recall/mAP values were not included in the supplied materials, this report identifies the missing information rather than fabricating measurements.

# 1. Introduction

Sports analysis increasingly depends on the ability to process large volumes of visual data. Manual review of football footage is time-consuming and can be inconsistent, particularly when many players appear simultaneously or when objects become partially occluded. An automated vision system can assist by locating players, referees, and other relevant objects frame by frame, while a tracking component can maintain identities across consecutive frames.

The AI Sports Analyst System addresses this need through a practical pipeline built around YOLOv8, DeepSORT, OpenCV, Google Colab, and Gradio. The project is important because it connects research methods in object detection and tracking to an accessible application that can be used for demonstrations, experimentation, and future sports-statistics development.

# 2. Problem Statement

The problem is to automatically identify and follow sports objects in football images and video despite changes in scale, camera position, background complexity, motion, and partial occlusion. A useful system must not only detect objects but also present the outputs in a way that is understandable to a user. The project therefore combines visual inference, tracking, result rendering, and an interactive dashboard.

# 3. Objectives

The principal objectives were to develop an AI-based sports-analysis prototype, train or apply a YOLO detector for relevant football classes, integrate DeepSORT to maintain object tracks, test the system on sports images and video, visualize confidence scores and bounding boxes, and expose the workflow through an interactive Gradio dashboard. A further objective was to document the system sufficiently to support later improvements such as live-stream processing and richer player statistics.

# 4. Related Work

Modern sports-analytics systems commonly combine object detection, multi-object tracking, and domain-specific analysis. YOLO-family detectors are widely used because they perform object localization and classification in a single inference pipeline, making them suitable for near-real-time applications [1]. DeepSORT extends tracking-by-detection by associating detections across frames using motion and appearance information [2]. OpenCV provides the image and video-processing functions required to read frames, manipulate visual data, and write annotated outputs [3].

The present project differs from a purely academic detector because it includes an end-user dashboard. Gradio supports rapid construction of interfaces around machine-learning functions, allowing a user to upload an image and inspect the resulting predictions [4]. Google Colab provides a convenient experimental environment for notebook-based development and GPU-assisted model execution [5]. Together, these components represent a practical integration of established computer-vision methods rather than a single isolated model.

| Component or approach | Role in existing work | Role in this project |
|---|---|---|
| YOLOv8 | Single-stage object detection and classification | Detects football-related objects and returns bounding boxes and confid |
| DeepSORT | Multi-object association over time | Maintains tracks across video frames after detection |
| OpenCV | Image, frame, and video processing | Supports input handling, frame operations, and annotated output |
| Gradio | Interactive machine-learning interfaces | Provides the AI Sports Analyst Dashboard |

# 5. Dataset

The supplied project outputs indicate a football-focused dataset with the classes **player**, **referee**, and **background**; the confusion-matrix visualization also displays a football label. The images show broadcast-style football scenes with multiple players, a referee, a pitch, stadium seating, advertising boards, and scoreboard overlays. These conditions are useful for testing object detection under realistic visual clutter.

**Dataset documentation note.** The provided materials do not state the exact number of images, the train/validation/test split, the source dataset name, or the number of annotated instances per class. These values should be added from the training notebook or dataset export before an official submission if they are available.

# 6. Data Annotation

The detection outputs imply an object-detection annotation format in which each relevant object is represented by a class label and a bounding box. The predicted visualizations use rectangular boxes around players and referees, with confidence values displayed beside the boxes. For a reproducible final report, the annotation subsection should record the annotation tool, label format, image-resolution policy, and quality-control procedure used to check missing or overlapping boxes.

# 7. Methodology

The complete pipeline begins with an image or video input. Frames are decoded and passed to YOLOv8, which predicts candidate objects, their classes, and confidence scores. Predictions below the selected confidence threshold can be filtered. The remaining detections are passed to DeepSORT, which estimates motion and associates detections between frames to produce stable track identities. OpenCV is then used to render boxes, labels, and tracking information on the output frames. Finally, the Gradio dashboard displays the input and detection result to the user.

| Stage | Input | Output |
|---|---|---|
| 1. Input handling | Sports image or video | Decoded image/frame data |
| 2. Detection | Frame | Class labels, bounding boxes, confidence scores |
| 3. Association | Consecutive detections | Track identities over time |
| 4. Visualization | Tracked detections | Annotated frame or saved video |
| 5. Interface | Annotated result | Dashboard display for user inspection |

# 8. Model

YOLOv8 is used as the detector. In the supplied images, the model produces class labels such as “player” and “referee” together with confidence scores and bounding boxes. DeepSORT is used after detection as the temporal association layer. This design is modular: the detector can be retrained or replaced without redesigning the user interface, while the tracker can be tuned independently for motion and appearance association.

**Training-configuration note.** The supplied materials do not include the exact YOLO model variant, number of epochs, batch size, image size, optimizer, learning rate, augmentation settings, or confidence/IoU thresholds. These settings should be copied from the training notebook into the final academic version if required by the instructor.

# 9. Evaluation

Evaluation was conducted through visual detection examples and a confusion matrix. The confusion matrix supplied with the project contains a dominant diagonal value of 101 for the player class, with smaller off-diagonal values including 4 and 3 in the player row, 5 and 1 in the referee-related region, and background-related values of 13 and 2. The matrix suggests that player detection is the strongest portion of the shown evaluation output, while confusion involving referee and background categories remains an area for improvement.

The required brief requests precision, recall, mAP, and a confusion matrix. The supplied images provide the confusion matrix but do not provide explicit precision, recall, or mAP tables. Those metrics should be inserted from the model-results file if available. It is preferable to state that the values are pending than to infer them from a rendered matrix.

```
from IPython.display import Image

Image(filename='/content/runs/detect/train/confusion_matrix.png', height=480)
```

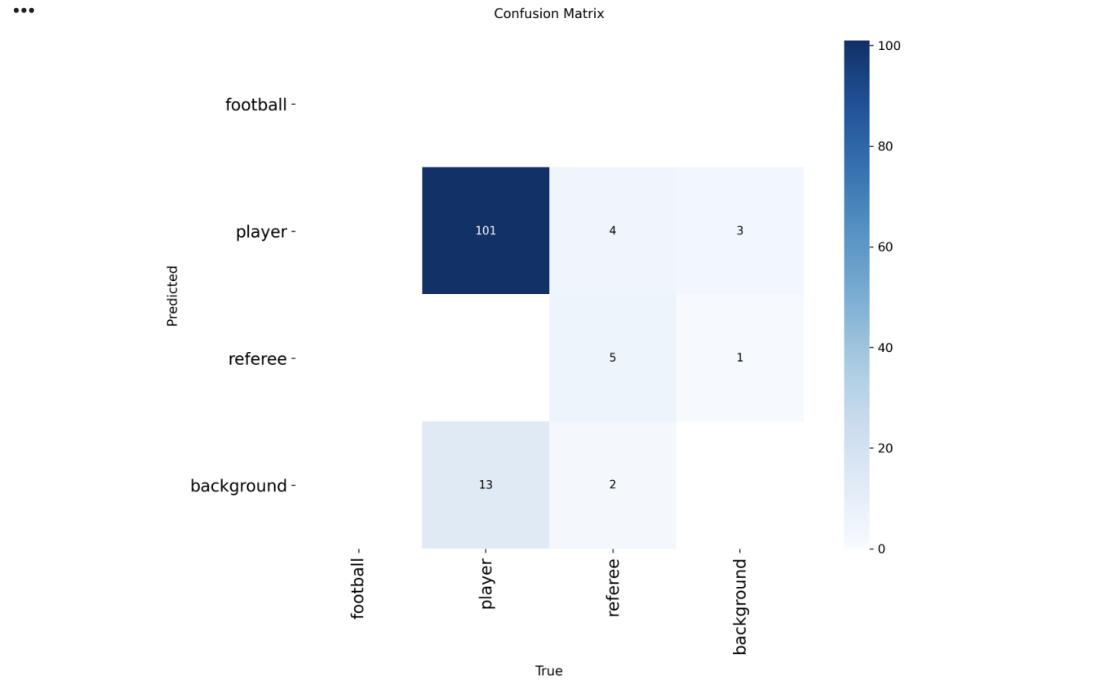



Figure 2. Supplied confusion matrix for the sports-object classes. The figure is reproduced as evidence of the evaluation output.

# 10. Results

The supplied results demonstrate that the system detects multiple players across different positions on the pitch and identifies a referee in at least one scene. Confidence scores are displayed next to the predicted labels, enabling a user to distinguish stronger detections from uncertain ones. The examples also show that the detector can operate in crowded scenes containing many nearby players.

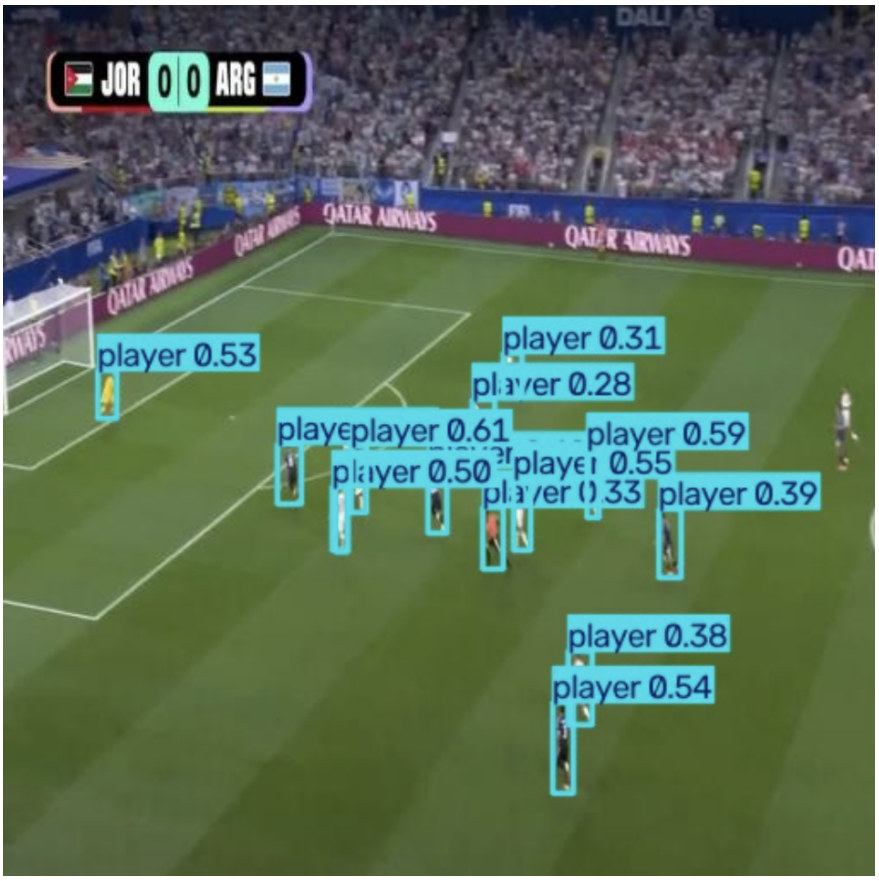



Figure 3. Example detection output with player predictions and confidence scores.

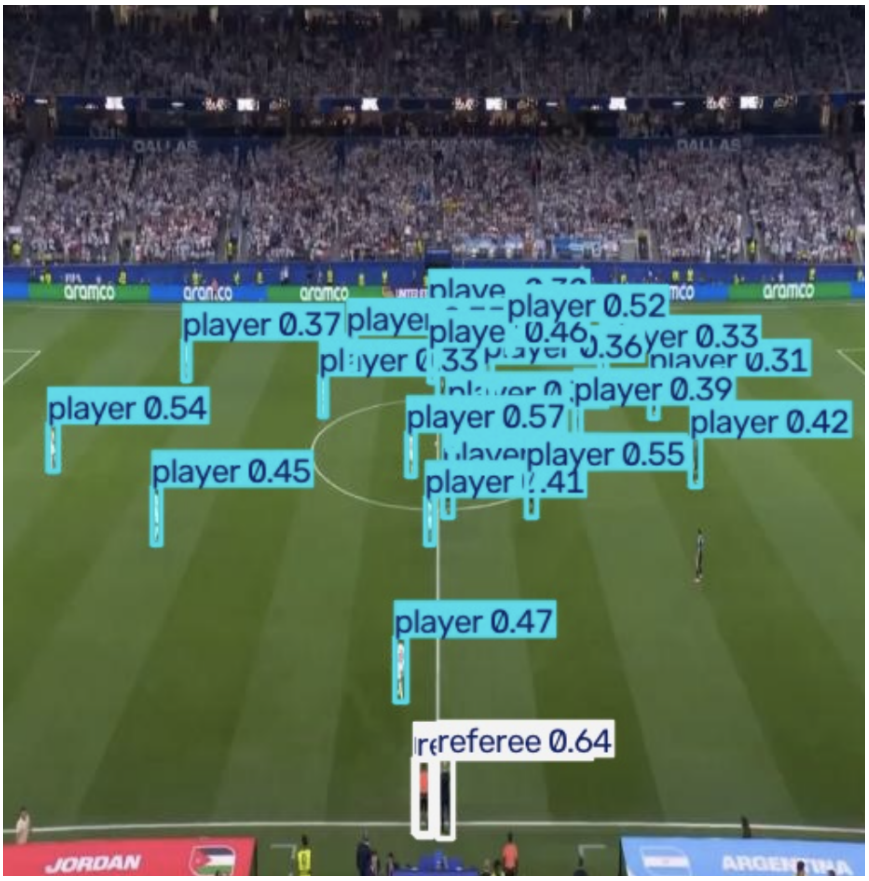



Figure 4. Example detection output showing many players and a referee near the lower portion of the frame.

Object Detection




Figure 5. Example object-detection result with player and referee labels rendered on a broadcast football frame.

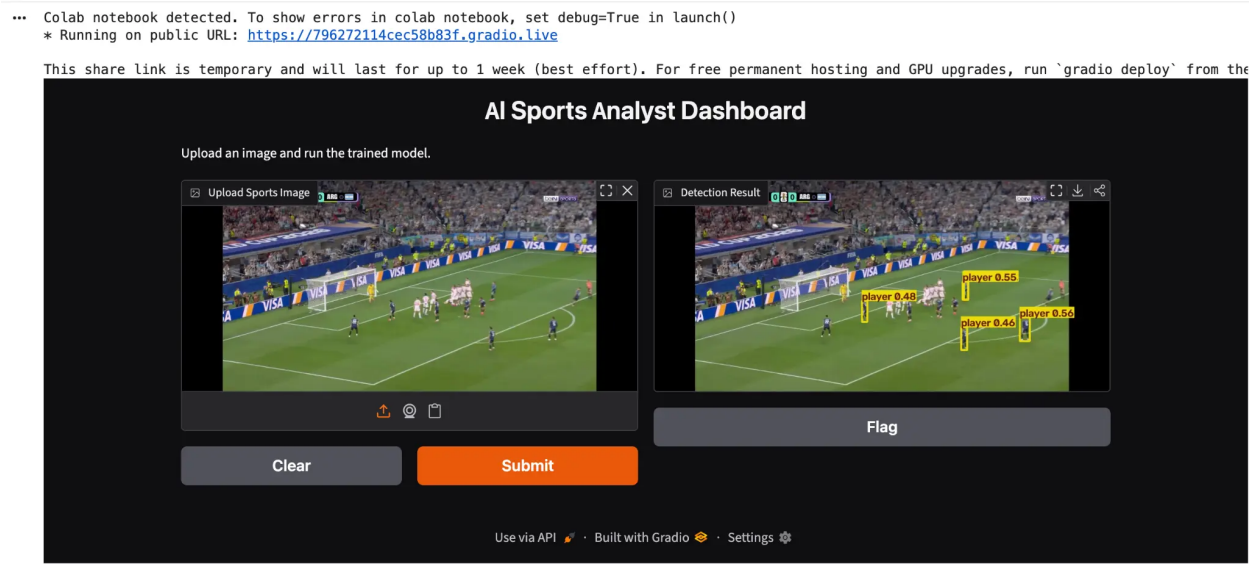



Figure 6. AI Sports Analyst Dashboard showing an uploaded sports image, the detection result, and interactive controls.

# 11. Limitations

The system has several limitations. Detection quality may decline when players overlap, become very small, are partially hidden, or have similar visual appearance. Broadcast camera motion, blur, compression artifacts, shadows, stadium advertising, and scoreboard graphics can also affect confidence. A tracker may lose an identity when a target disappears for several frames or when two players cross paths. In addition, the current dashboard demonstrates inference and visualization but does not yet provide a complete statistical engine for passes, possession, formations, distance covered, or tactical events.

The evaluation evidence is also incomplete because the supplied materials do not contain the full dataset specification or the numerical metric report. Consequently, the shown examples support a qualitative assessment, but they should not be presented as a complete benchmark without the underlying test-set statistics.

# 12. Future Work

Future improvements include real-time live-stream processing, incorporation of advanced player statistics, stronger tracking algorithms, and deployment as a cloud application. Real-time processing would require a stable frame-ingestion service and latency-aware inference. Advanced statistics could be built by mapping image coordinates to the pitch and aggregating tracks into events such as movement, possession, and team structure. Stronger tracking could reduce identity switches through improved appearance embeddings, camera-motion compensation, and more robust occlusion handling. Cloud deployment would make the system accessible through a scalable web service, subject to compute cost, privacy, and video-storage considerations.

# Video Demonstration

Scan the QR Code below to open the project video demonstration. The QR Code encodes the Google Drive link supplied by the project owner. Access will depend on the Google Drive sharing setting; the file should be configured as “Anyone with the link can view.”



Figure 7. QR Code for the project video demonstration.

https://drive.google.com/file/d/1VEgM0C02bUvYoD0o0sZ3LHH81_kJW3YG/view?usp=sharing

# 13. Conclusion

This project developed an AI Sports Analyst System capable of detecting and tracking sports objects using YOLOv8 and DeepSORT. The system successfully analyzed sports images and video outputs, generated bounding-box predictions and confidence scores, and provided an interactive dashboard for user interaction. The supplied confusion matrix and detection examples demonstrate the practical usefulness of the pipeline, particularly for identifying players and referees in football scenes. The project also establishes a foundation for real-time analysis, richer player statistics, stronger tracking, and cloud-based deployment.

# 14. References


[1] Ultralytics. *Ultralytics YOLO Documentation*. https://docs.ultralytics.com

[2] levan92. *deep_sort_realtime Repository*. https://github.com/levan92/deep_sort_realtime

[3] OpenCV. *OpenCV Documentation*. https://opencv.org

[4] Gradio. *Gradio Documentation*. https://www.gradio.app

[5] Google. *Google Colab*. https://colab.research.google.com

[6] Jocher, G., et al. *YOLO Research Resources*. Available through the Ultralytics documentation and associated research references.


**Note on supplied evidence.** Figures in this report were provided by the project owner. They are included to document the observed model outputs and interface, not as independently reproduced experiments.